DEPI 2026 · AI & DATA SCIENCE · MACHINE LEARNING

AI-Powered Brain MRI Tumor Analysis

Segmentation, volumes, change over time, radiomics and a confidence map, from four MRI scans

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Instructor: Eng. Aya Abdallah · Decision-support research prototype, not a diagnostic tool

THE PROBLEM

Outlining a brain tumor by hand is slow and varies between readers

3D, four scans

Each patient has T1, T1ce, T2 and FLAIR volumes of about 155 slices. Gliomas have three nested parts to outline.

Change over time

Treatment decisions depend on whether the tumor grows or shrinks between scans, which needs consistent volumes.

Our goal

An automatic, measurable second reader that also says where it is unsure, for decision support only.

WHAT THE SYSTEM DOES

Four scans in, a measured tumor out

Input

T1, T1ce, T2, FLAIR (NIfTI), one or more visits

→

Segment

nnU-Net finds whole tumor, tumor core and enhancing tumor in 3D

→

Measure

Volumes, change between visits, radiomics features

→

Explain

Confidence map showing where the model is unsure

Served by a FastAPI backend to a web interface. Every output carries a decision-support notice.

DATA

Five public cohorts, with the test patients locked

Cohort	Use	Patients	Train / val / test
BraTS 2021	Training and validation	1,251	875 / 188 / 188
MU-Glioma-Post	Change over time	203	142 / 31 / 30
RHUH-GBM	Change over time	40	28 / 6 / 6
BraTS-Africa	External test	146	locked
UTSW-Glioma	External test	625	locked

Every file is checksummed (SHA-256). The code refuses to load locked patients before the final test. Cohorts that overlap BraTS 2021 (UPenn-GBM, UCSF-PDGM, TCGA) were rejected as external tests.

SEGMENTATION MODEL

nnU-Net beat our 3D U-Net baseline by 0.14 mean Dice

Model (188 validation patients)	TC	WT	ET	Mean
3D U-Net baseline (EXP-0001)	0.717	0.848	0.728	0.765
nnU-Net 3d_fullres: seg-model v1	**0.917**	**0.935**	**0.872**	**0.908**

Dice measures overlap with the expert mask (1 = perfect). TC = tumor core, WT = whole tumor, ET = enhancing tumor. seg-model v1 is registered with a model card and a SHA-256 checksum. These are validation scores, not clinical performance.

QUANTITATIVE ANALYSIS

One command gives every measurement for a patient

Volumes

cm³ of WT, TC, ET and of each label: necrosis, edema, enhancing tumor, resection cavity

Change over time

cm³ and % between every pair of visits; % only when the earlier volume is at least 1 cm³

Radiomics

Shape, intensity and texture features per region (PyRadiomics), with a quality note

Example: RHUH-0002

3 visits in 100 s. Pre-op Dice vs expert: WT 0.948, TC 0.777, ET 0.740

CONFIDENCE MAP

The map points at the model’s own mistakes

0.89–0.94

AUROC for telling wrong voxels from right ones, per region

39–47%

of wrong voxels caught by flagging only 1% of brain voxels

−0.68 to −0.81

correlation between a case’s uncertainty and its Dice

Built from nnU-Net’s 8 mirror flips, run one by one and compared. The mask stays exactly seg-model v1’s. Measured on the 188 validation patients.

FAST PREDICTION

A tumor mask in 2.1 seconds per scan

T4 GPU, 20 validation patients	Time per scan	Mask vs seg-model v1	Mean Dice vs expert
Original nnU-Net	8.7 s	–	0.913
Export, 8 mirror flips	8.6 s	identical (0 voxels differ)	0.913
Export, single pass	**2.1 s**	about 2 cm³ differ, at edges	0.911

The network is exported to TorchScript and checked by SHA-256. Loading takes 1.0 s instead of 4.9 s, with no 250 MB training checkpoint. The FastAPI backend serves it: /health, /predict, /predict/longitudinal.

HOW IT IS BUILT

Open code, shared data, a registered model

Code

Python package bmts on GitHub; forks and pull requests; pytest suite (66 tests)

Data and compute

Shared Google Drive and Colab T4 GPUs; no data or weights in the repository

Tracking

MLflow experiments, model registry with model card, Notion task board

Backend

FastAPI: validates uploads, runs the model, returns volumes and changes as JSON

Interface

Streamlit slice viewer done; web interface in progress

Deployment

Docker and Azure planned for the next step

PROJECT PLAN

Five milestones in six weeks, 30 tasks

Milestone	Dates	Status on 31 Oct
M1 Data foundation	23 Sep – 4 Oct	Done
M2 Segmentation model	26 Sep – 9 Oct	Done
M3 Quantitative analysis	10 – 18 Oct	Done
M4 Application and deployment	17 – 28 Oct	In progress: model and API done
M5 Final validation and delivery	15 – 31 Oct	Not started

Each task has one owner, a deadline and a “done when” check; each milestone ends with a review task. Team: Mohamed Tagy, Alaa Emad, Hagar Aiman, Yasmin Mahmoud, Maryam Ahmed, Essam Mohamed.

What we can claim, and what comes next

Limits

- Validation scores only; locked test not yet run
- Needs cleaned, skull-stripped BraTS-style scans
- Early post-surgery scans are harder
- Not a medical device; expert review required

Next steps

- Finish the web interface
- Docker image and Azure deployment
- Final test on locked and external patients
- Final report and reproducibility package

Thank you

Code and documentation: github.com/nhahub/NHA-5-187

Decision-support research prototype. It does not diagnose, and it has not been validated for clinical use.